

MANUFACTURING ERP

Product Requirements Document • Technical Specification

Version	1.0 — MVP
Document Type	Technical Specification
Modules	Sales, Inventory, BOM, Production, MES, Shipping, Dashboard, RBAC
Status	Draft — For Review

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1. System Overview

The Manufacturing ERP provides end-to-end visibility from customer order placement through manufacturing execution and final delivery. The system integrates eight core modules that work in concert to ensure accurate inventory tracking, efficient production scheduling, and timely shipment fulfillment.

1.1 Core Business Flow



1.2 Module Summary

Module	Responsibility
Sales Orders	Manage customer orders from draft to delivery; track order status and shipment.
Inventory	Track raw materials and finished goods; manage stock levels and adjustments.
Bill of Materials (BOM)	Define components required per product; support multi-level BOM structures.
Production Orders	Plan and execute manufacturing jobs; link to BOM and raw materials.
Manufacturing Execution	Record actual production; update inventory upon completion.
Shipping	Manage packing, dispatch, and delivery of finished goods to customers.
Dashboard & Analytics	KPIs, charts, and reports for business intelligence and decision-making.
RBAC	Role-based access control to secure module-level permissions per user role.

2. User Roles & Permissions (RBAC)

The system implements Role-Based Access Control (RBAC) with five distinct roles. Each role grants specific module-level permissions based on the user's function within the organization.

Role	Permissions
Admin	Full system access
	User management
	Role management
	Audit log access
	All CRUD operations across all modules
Sales Executive	Create/manage Customers
	Create/manage Sales Orders
	Track Shipments
	Read-only Inventory access
Production Manager	BOM management (create/edit/archive)
	Production Order management
	Manufacturing execution
	Inventory visibility (read)
Inventory Manager	Raw material management
	Stock adjustments
	Goods receipt processing
	Warehouse management
Viewer	Read-only access to Dashboards
	Read-only access to Reports

3. Sales Order Module

3.1 Status Definitions

Status	Type	Description
DRAFT	Normal	Order created but not yet submitted. Can be edited or deleted.
SUBMITTED	Normal	Order submitted by Sales Executive; awaiting inventory check and approval.
INVENTORY CHECK	Normal	System is verifying if sufficient finished goods stock exists to fulfill the order.
APPROVED	Normal	Order approved after confirming stock availability or accepting partial fulfillment.
IN PRODUCTION	Normal	A Production Order has been created to manufacture required quantities.
READY TO SHIP	Normal	All ordered items are available in Finished Goods inventory and ready for packing.
SHIPPED	Normal	Order has been dispatched to the customer; shipment is in transit.
DELIVERED	Terminal	Order successfully delivered to the customer. Final state.
ON HOLD	Exception	Order paused due to payment issues, customer request, or resource constraints.
CANCELLED	Exception	Order cancelled by Sales Executive or Admin. No further processing.
REJECTED	Exception	Order rejected due to invalid data, credit failure, or policy violation.

3.2 Status Transition Flow

#	From Status	→	To Status	Trigger / Condition
1	DRAFT	→	SUBMITTED	Sales Executive submits the order
2	SUBMITTED	→	INVENTORY CHECK	System triggers automatic inventory validation
3	INVENTORY CHECK	→	APPROVED	Stock >= Order Quantity (no production needed)
4	INVENTORY CHECK	→	IN PRODUCTION	Stock < Order Quantity → Production Order created
5	IN PRODUCTION	→	APPROVED	Production Order completes; finished goods updated
6	APPROVED	→	READY TO SHIP	All items confirmed available in finished goods
7	READY TO SHIP	→	SHIPPED	Shipment dispatched (Dispatched status in Shipment module)
8	SHIPPED	→	DELIVERED	Delivery confirmed by logistics or customer
9	Any Active Status	→	ON HOLD	Admin or authorized user places order on hold

#	From Status	→	To Status	Trigger / Condition
10	Any Active Status	→	CANCELLED	Cancellation request by Sales Executive or Admin
11	SUBMITTED	→	REJECTED	Validation failure or policy violation on review

3.3 Inventory Check Logic — Worked Example

When a Sales Order is approved, the system automatically checks Finished Goods inventory:

Scenario	Condition	System Action
Customer orders 100 Chairs	Finished Goods Stock = 100 (Stock >= Order)	→ Status: Ready To Ship
Customer orders 100 Chairs	Finished Goods Stock = 30 (Stock < Order)	Shortage = 70. Create Production Order for 70 Chairs

4. Inventory Module

4.1 Inventory Types

Type	Description	Managed By
Raw Materials	Components and inputs for production	Inventory Manager
Finished Goods	Completed products ready for sale/shipment	Production Manager / Inventory Manager
WIP (Work-in-Progress)	Materials in active production process	Production Manager

4.2 Inventory Movement Types

Movement Type	Description
Material Consumption	Raw materials consumed during production execution
Finished Goods Receipt	Finished goods added to stock upon production completion
Goods Receipt (PO)	Raw materials received from supplier purchase orders
Stock Adjustment	Manual corrections to inventory (damage, count correction, etc.)
Sales Order Reservation	Quantity reserved against a confirmed sales order
Shipment Issue	Finished goods removed from stock when order is shipped
Return Receipt	Goods returned by customer added back to inventory

4.3 Low Stock Alert

Each raw material can be configured with a Minimum Stock Level (Reorder Point). When on-hand quantity falls below this threshold, the material appears on the Dashboard under 'Low Stock Materials' and may trigger a purchase requisition.

5. Bill of Materials (BOM) Module

A BOM defines the list of raw materials and their quantities required to manufacture one unit of a finished product. The system uses BOM data to calculate material requirements when a Production Order is created.

5.1 BOM Status Definitions

Status	Type	Description
DRAFT	Normal	BOM is being created or edited. Cannot be used in production orders.
ACTIVE	Normal	BOM is approved and available for use in Production Orders.
ARCHIVED	Terminal	BOM is retired. Historical reference only; cannot be used in new production orders.

5.2 BOM Worked Example — Chair

The following example illustrates a BOM for a standard Chair product:

Component / Raw Material	Qty per 1 Chair	Qty for 70 Chairs	Qty for 100 Chairs
Wood (planks)	5	350	500
Screws	20	1,400	2,000
Paint (litres)	1	70	100

Formula: *Required Quantity = BOM Quantity per Unit × Production Order Quantity*

5.3 BOM Transition Rules

- DRAFT → ACTIVE: BOM reviewed and approved by Production Manager
- ACTIVE → ARCHIVED: New version supersedes the current BOM, or product is discontinued
- DRAFT → (deleted): Incomplete BOMs may be discarded without archiving

6. Production Order Module

6.1 Status Definitions

Status	Type	Description
PLANNED	Normal	Production Order created from a Sales Order shortfall. Materials not yet reserved.
RELEASED	Normal	Order officially released to the shop floor. Production team is notified.
MATERIAL RESERVED	Normal	Required raw materials have been reserved/allocated from inventory for this order.
IN PROGRESS	Normal	Manufacturing has started. Raw materials are being consumed.
QUALITY CHECK	Normal	Production complete; finished goods undergoing quality inspection before acceptance.
COMPLETED	Terminal	Quality passed. Finished goods added to inventory. Sales Order updated.
ON HOLD	Exception	Production paused due to material shortage, equipment issue, or management decision.
CANCELLED	Exception	Production Order cancelled. Reserved materials returned to available inventory.
REJECTED	Exception	Quality check failed. Order flagged for rework or disposal.

6.2 Status Transition Flow

#	From Status	→	To Status	Trigger / Condition
1	PLANNED	→	RELEASED	Production Manager reviews and releases the order to shop floor
2	RELEASED	→	MATERIAL RESERVED	System reserves raw materials per BOM calculation
3	MATERIAL RESERVED	→	IN PROGRESS	Operator confirms production start in MES
4	IN PROGRESS	→	QUALITY CHECK	Production execution completed; quantity produced recorded
5	QUALITY CHECK	→	COMPLETED	Quality inspection passed; finished goods receipted to inventory
6	QUALITY CHECK	→	REJECTED	Quality inspection failed; order flagged for investigation
7	Any Active Status	→	ON HOLD	Production Manager or Admin places order on hold

#	From Status	→	To Status	Trigger / Condition
8	Any Active Status	→	CANCELLED	Cancellation authorized; reserved stock released

6.3 Trigger Condition

A Production Order is automatically created when a Sales Order's Inventory Check detects a shortage:

- Shortage Quantity = Ordered Quantity – Available Finished Goods Stock
- The Production Order quantity equals the shortage quantity
- BOM is looked up to calculate raw material requirements for that quantity

7. Manufacturing Execution System (MES)

Upon completion of a Production Order, the MES module performs the following inventory transactions automatically:

Action	Inventory Effect	Movement Type Recorded
Production Execution	Raw Materials decrease by BOM quantity × units produced	Material Consumption
Production Completion	Finished Goods increase by units produced	Finished Goods Receipt
Adjustment (optional)	Manual correction for scrap, rework, or yield variance	Adjustment Entry

7.1 MES Worked Example

Production Order: 70 Chairs

Material	Before Production	Consumed	After Production
Wood (planks)	500	– 350	150
Screws	2,000	– 1,400	600
Paint (litres)	100	– 70	30
Chairs (Finished Goods)	30	+ 70	100

8. Shipment Module

8.1 Status Definitions

Status	Type	Description
READY TO SHIP	Normal	Finished goods available; shipment can be initiated.
PACKING	Normal	Items being physically packed and labelled in the warehouse.
DISPATCHED	Normal	Shipment picked up by courier / carrier. Inventory deducted.
IN TRANSIT	Normal	Shipment is en route to customer destination.
DELIVERED	Terminal	Customer has received the shipment. Sales Order marked Delivered.
RETURNED	Exception	Customer returned goods. Return receipt created; inventory updated.
DAMAGED	Exception	Goods damaged in transit. Damage report created; replacement may be issued.
CANCELLED	Exception	Shipment cancelled before dispatch. Inventory reservation released.

8.2 Shipment Flow

#	From Status	→	To Status	Trigger / Condition
1	READY TO SHIP	→	PACKING	Warehouse staff initiates packing process
2	PACKING	→	DISPATCHED	Carrier collects shipment; tracking number assigned
3	DISPATCHED	→	IN TRANSIT	Carrier confirms goods in transit
4	IN TRANSIT	→	DELIVERED	Delivery confirmation received
5	DELIVERED	→	RETURNED	Customer initiates return within return window
6	IN TRANSIT / DELIVERED	→	DAMAGED	Damage report filed by customer or carrier
7	PACKING / READY TO SHIP	→	CANCELLED	Shipment cancelled before dispatch

9. Dashboard & Analytics

The Dashboard provides real-time KPIs and visual analytics accessible to all roles (filtered by permissions). Data refreshes automatically and can be filtered by date range, product, customer, or warehouse.

KPI / Widget	Description
Revenue (MTD/YTD)	Total sales revenue for the current month and year
Total Orders	Count of Sales Orders by status (Active, Completed, Cancelled)
Production Orders	Count of Production Orders by status
Inventory Value	Total valuation of raw materials + finished goods stock
Low Stock Materials	Raw materials at or below reorder point — requires immediate action
Pending Shipments	Shipments in Packing, Dispatched, or In Transit status
Top Customers	Customers ranked by order value in the selected period
Top Products	Products ranked by units sold or revenue in the selected period

10. Audit Trail & Tracking

All records in the system maintain full audit metadata. This ensures traceability and accountability across all operations.

10.1 Standard Audit Fields

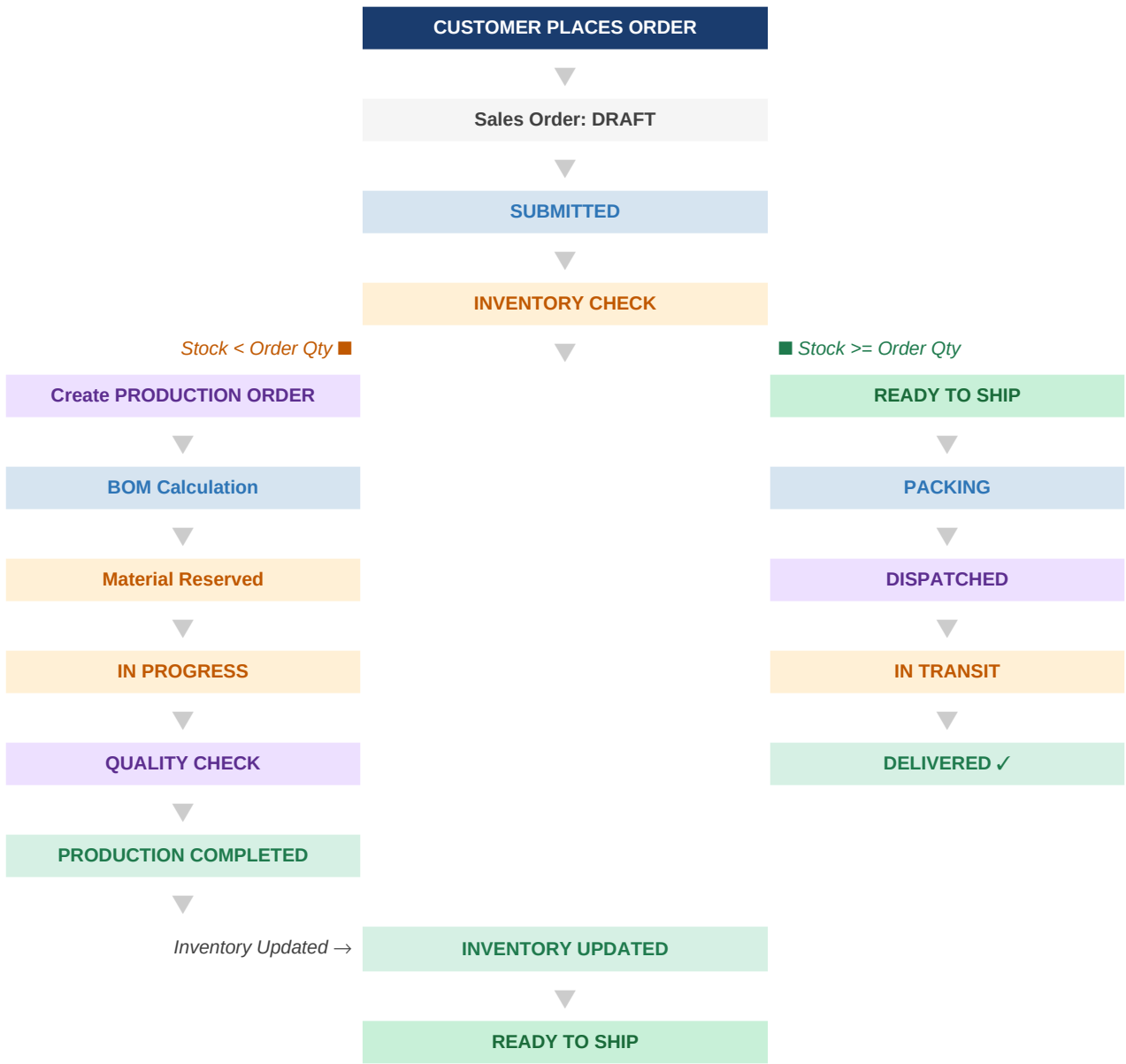
Field	Description	Applies To
Created By	User who created the record	All modules
Created Date	Timestamp of record creation	All modules
Updated By	User who last modified the record	All modules
Updated Date	Timestamp of last modification	All modules
Status Changed By	User who triggered the last status change	Orders, Shipments, Production
Status Changed At	Timestamp of the last status transition	Orders, Shipments, Production

10.2 Audit Log Events

- Sales Order status changes and modifications
- Inventory adjustments and stock movements
- Production Order status changes
- BOM create/edit/archive events
- Shipment dispatches and deliveries
- User management (create, deactivate, role assignment)
- Login and logout events (Admin audit only)

11. End-to-End System Flowchart

The following diagram illustrates the complete order-to-delivery flow, including both the in-stock and production paths:



11.1 Flow Legend

Dark Blue	System entry / customer interaction	Purple	Production / quality steps
Orange	Decision / material steps	Green	Completion / success states

12. Master Data

The following master data entities form the foundation for all transactional modules:

Entity	Key Attributes
Customers	Customer ID, Name, Contact, Address, Credit Limit, Payment Terms
Products (Finished Goods)	Product ID, Name, SKU, Unit of Measure, BOM Reference, Price
Raw Materials	Material ID, Name, Unit, Reorder Point, Supplier(s), Current Stock
Suppliers	Supplier ID, Name, Contact, Lead Time, Default Payment Terms
Warehouses	Warehouse ID, Name, Location, Capacity, Zones
Users	User ID, Name, Email, Role, Status (Active/Inactive), Last Login
Roles & Permissions	Role Name, Module Access List, Create/Read/Update/Delete flags